

DODGE CENTER, MN, USA

WMO: 726596

Lat: 44.018N

Lon: 92.831W

Elev: 398

StdP: 96.64

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-22.1	-27.8	0.3	-23.3	-25.9	0.4	-21.6	12.9	-6.1	11.9	-6.4	3.9	300	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.2	23.3	29.0	22.2	27.7	21.2	25.1	28.6	24.0	27.7	22.9	26.4	6.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	19.7	27.5	22.6	18.2	26.4	22.1	17.6	25.9	78.7	28.4	74.5	27.6	69.7	27.0	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	10.8	9.3	-28.1	33.7	3.2	1.7	-30.4	34.9	-32.2	35.9	-34.0	36.9	-36.4	38.1
			-28.3	27.0	3.2	1.5	-30.6	28.1	-32.4	29.0	-34.2	29.8	-36.5	30.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.1	-8.1	-0.1	7.6	14.1	19.7	21.4	20.0	16.6	8.9	1.6	-6.1	(6)
	DBStd	12.16	7.21	6.56	6.99	5.70	5.03	3.49	3.00	3.08	4.40	5.33	6.08	6.46	(7)
	HDD10.0	2396	592	506	322	111	18	0	0	0	4	87	258	498	(8)
	HDD18.3	4323	851	739	572	323	149	24	7	18	84	297	502	756	(9)
	CDD10.0	1408	0	0	8	38	146	292	354	310	202	51	7	0	(10)
	CDD18.3	296	0	0	1	1	19	66	103	69	32	4	0	0	(11)
	CDH23.3	2375	0	0	2	27	197	534	839	487	252	37	0	0	(12)
	CDH26.7	602	0	0	0	6	59	151	220	102	57	6	0	0	(13)
(14) Wind	WSAvg	4.4	4.9	4.9	4.9	5.2	4.9	4.0	3.3	2.8	3.7	4.3	4.7	4.8	(14)
(15) Precipitation	PrecAvg	883	25	21	51	85	101	138	108	127	92	60	47	28	(15)
	PrecMax	1051	60	41	103	163	198	261	216	319	300	158	165	69	(16)
	PrecMin	587	9	3	15	26	28	65	27	24	26	16	4	5	(17)
	PrecStd	134	15	11	25	40	43	59	52	60	64	37	37	16	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.2	22.0	27.3	32.1	32.7	32.5	31.3	31.3	27.5	20.0	11.1	(19)
		MCWB	4.5	5.2	15.0	17.9	19.8	22.8	25.3	24.6	21.8	18.2	14.1	8.2	(20)
	2%	DB	3.1	4.0	16.3	22.4	27.8	30.2	31.1	28.9	27.9	23.0	16.1	6.3	(21)
		MCWB	1.5	2.6	12.6	14.3	17.7	21.9	24.9	23.6	20.8	16.4	11.8	5.0	(22)
	5%	DB	2.2	2.6	12.6	19.1	25.9	27.9	28.9	27.6	26.4	21.0	12.7	3.6	(23)
		MCWB	1.2	1.5	9.8	12.7	17.1	20.2	23.5	22.3	19.6	15.5	9.5	2.6	(24)
	10%	DB	1.0	1.2	8.9	17.2	22.5	26.4	27.5	26.2	24.0	17.4	10.1	2.3	(25)
		MCWB	0.3	0.4	7.1	11.0	15.4	19.4	22.2	21.1	18.8	12.5	7.6	1.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	5.5	16.4	19.2	22.8	25.6	27.9	25.8	24.0	21.6	15.5	9.1	(27)
		MCDB	6.3	6.6	20.2	24.2	27.4	30.0	31.1	28.8	28.4	24.7	18.8	9.9	(28)
	2%	WB	2.3	3.2	13.2	15.8	20.1	23.8	25.7	24.4	22.4	18.0	12.6	5.0	(29)
		MCDB	3.5	4.2	15.5	20.3	24.8	28.1	28.9	27.9	26.3	21.9	15.1	6.5	(30)
	5%	WB	1.1	1.7	10.0	13.8	18.7	22.5	24.4	23.3	21.1	15.3	10.1	2.8	(31)
		MCDB	1.8	2.4	12.8	18.0	23.0	26.2	27.9	26.6	25.0	18.7	13.0	3.5	(32)
	10%	WB	-0.1	0.2	7.1	11.9	17.1	21.1	23.2	22.1	19.6	13.2	7.6	1.3	(33)
		MCDB	0.7	1.1	9.0	15.9	20.8	24.8	26.6	25.2	23.5	17.1	9.5	1.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	8.4	10.4	10.8	10.1	10.2	10.3	11.2	10.2	8.6	7.3	(35)
		MCDBR	9.1	8.8	12.6	15.1	14.2	12.4	11.1	11.1	12.9	14.3	12.6	8.7	(36)
	5% WB	MCWBR	7.9	7.8	8.9	9.0	6.7	6.1	6.3	6.5	6.9	7.9	8.5	7.7	(37)
		MCWBR	8.4	8.4	11.9	13.3	11.5	10.5	10.3	10.0	11.0	11.5	12.0	8.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.263	0.289	0.313	0.364	0.391	0.402	0.401	0.397	0.366	0.321	0.287	0.263	(40)
		taud	2.425	2.338	2.405	2.316	2.293	2.303	2.303	2.322	2.381	2.497	2.510	2.451	(41)
	Ebn at Noon		879	905	928	899	880	869	864	856	857	861	844	843	(42)
		Edh at Noon	72	95	102	122	128	128	126	120	104	81	67	64	(43)
(44) All-Sky Solar Radiation	RadAvg		1.76	2.82	3.75	4.58	5.37	6.00	6.29	5.30	4.21	2.71	1.77	1.29	(44)
	RadStd		0.20	0.30	0.37	0.48	0.42	0.47	0.26	0.36	0.30	0.36	0.16	0.11	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page